HU Collection

Engineering and Science Admissions Test

Mock Set F

Physics

Time allowed: 40 minutes
Questions: 27 multiple choice
Calculators: not permitted
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Date written: 29 October 2026

Name: Date sat: Score: / 27

Instructions

Answer all 27 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers, so answer every question. Calculators are not permitted. Take the gravitational field strength g to be 10 N kg^{-1} unless a question says otherwise. This paper is deliberately longer than the time allows: move on from any question that is costing you more than two minutes, and come back to it at the end.

Q	1	2	3	4	5	6	7	8	9	10
Ans										

Q	11	12	13	14	15	16	17	18	19	20
Ans										

Q	21	22	23	24	25	26	27			
Ans										

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1. Small aquarium air pumps contain a springy arm that is made to vibrate by an electromagnet. In one test rig, the electromagnet is a coil of many turns with no core, connected to the 50 Hz a.c. mains. Two designs of arm are compared. In design X, the end of the arm nearest the coil carries a small block of soft iron. In design Y, it carries instead a small permanent bar magnet, lying along the axis of the coil. The coil does not change the magnetisation of this magnet. In both designs the arm never touches the coil, and the block or magnet stays close to one end of the coil throughout.

Which of the following statements must be true?

I. In design X, the magnetic force on the soft iron block reaches its greatest magnitude 100 times each second.

II. In design Y, the magnetic force on the magnet reaches its greatest magnitude in the direction towards the coil 100 times each second.

III. In design Y, the magnetic force on the magnet changes direction 100 times each second.

A II only

B II and III only

C I only

D III only

E I and II only

F I and III only

G I, II and III

H none of them

2. Two small balls are joined by a light spring that obeys Hooke's law. Ball P has mass 0.20 kg and ball Q has mass 0.60 kg. A student holds P still, so that the spring hangs vertically with Q at rest below it. She then lets go of P. Ignore air resistance and take $g = 10$ N kg^{-1}.

Which of the following statements is/are correct?

I. Immediately after P is released, the acceleration of Q is zero.

II. Immediately after P is released, the acceleration of P is 30 m s^{-2}.

III. For as long as the balls are falling and have not touched each other, the total momentum of P and Q increases by 8.0 kg m s^{-1} every second.

A III only

B I only

C none of them

D II only

E I and III only

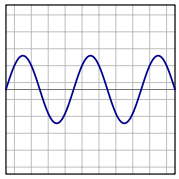
F II and III only

G I and II only

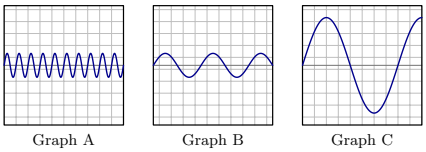
H I, II and III

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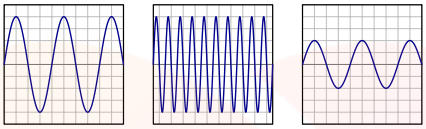
3. A microphone connected to an oscilloscope picks up a steady note from a loudspeaker. The output p.d. of the microphone is proportional to the variation in air pressure at the microphone. With each horizontal division of the screen representing 1.0 ms and each vertical division representing 2.0 mV, the trace is as shown on the upper screen.



upper screen: 1.0 ms and 2.0 mV per division



Graph A Graph B Graph C



Graph D Graph E Graph F

The note is then raised by two octaves (each octave doubles the frequency), and it is also made quieter, so that the amplitude of the pressure variation at the microphone is halved. At the same time the oscilloscope is reset so that each horizontal division represents 0.25 ms and each vertical division represents 0.50 mV.

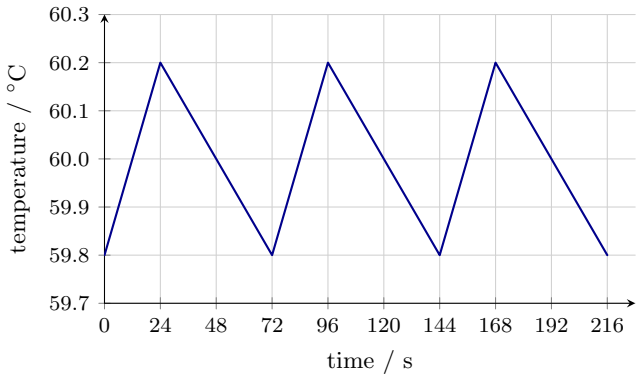
All seven screens have the same grid of divisions. Which of the graphs A to F shows the new trace?

A Graph A **B** Graph B **C** Graph C

D Graph D **E** Graph E **F** Graph F

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4. A sous-vide cooker keeps a bath of water close to 60°C. Its heater is either fully on, supplying energy to the water at 630 W, or off. A thermostat switches the heater off when the water reaches 60.2°C and on again when the water has cooled to 59.8°C. The graph shows the temperature of the water over three cycles.



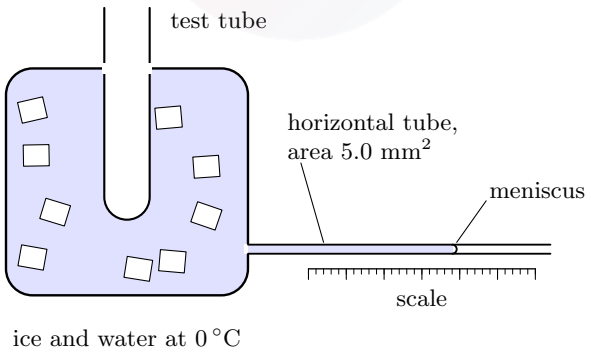
The water is stirred so that its temperature is uniform, and the heat capacity of the container is negligible. The bath loses energy to the surroundings at a rate that may be treated as constant over this narrow range of temperature. The specific heat capacity of water is 4200 J kg^{-1} K^{-1}.

What is the mass of water in the bath?

A 9.0 kg **B** 3.0 kg **C** 6.0 kg

D 12 kg **E** 27 kg **F** 18 kg

5. An ice calorimeter measures energy by using the fact that ice shrinks when it melts. A sealed glass vessel is completely filled with a mixture of ice and water at 0 °C, and is itself packed in ice at 0 °C, so that no energy passes between the vessel and its surroundings. A thin-walled test tube projects down into the vessel. A narrow horizontal glass tube, of uniform internal cross-sectional area 5.0 mm^2, leads out of the vessel; the water fills this tube up to a meniscus, whose position is read against a scale.



A metal sample of mass 66 g at 100 °C is dropped into the test tube. It cools to 0 °C, and all the energy it gives out melts some of the ice in the vessel. As a result the meniscus moves 20 cm along the tube.

Take the density of ice to be 0.90 g cm^{-3} and the density of water to be 1.0 g cm^{-3}. The specific latent heat of fusion of ice is 3.3×10^5 J kg^{-1}.

What is the specific heat capacity of the metal, and in which direction does the meniscus move?

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A 405 J kg^{-1} $°C^{-1}$; towards the vessel

B 500 J kg^{-1} $°C^{-1}$; towards the vessel

C 450 J kg^{-1} $°C^{-1}$; away from the vessel

D 45 J kg^{-1} $°C^{-1}$; towards the vessel

E 450 J kg^{-1} $°C^{-1}$; towards the vessel

F 500 J kg^{-1} $°C^{-1}$; away from the vessel

G 4500 J kg^{-1} $°C^{-1}$; towards the vessel

6. Battery capacities are often given in milliamp hours (mAh): 1 mAh is the charge that flows when a current of 1 mA flows for 1 hour. A power bank contains cells that can deliver a total charge of 10 000 mAh at a constant p.d. of 3.6 V. A converter inside the power bank supplies its USB output at 5.0 V, with an efficiency of 75%.

A phone is charged from this output through a USB cable of negligible resistance. The phone's battery holds 3000 mAh when full, and it is charged at a constant p.d. of 3.6 V by a charging circuit inside the phone. This circuit takes its input from the USB cable at 5.0 V and has an efficiency of 80%.

Starting with the power bank full, the phone's battery is charged from empty to full again and again. What charge passes along the USB cable during each full charge of the phone, and how many complete charges from empty to full can the power bank provide?

A 3000 mAh; 3

B 2700 mAh; 3

C 2700 mAh; 2

D 3750 mAh; 2

E 3000 mAh; 2

F 2160 mAh; 2

G 3750 mAh; 1

7. A winch is driven by a d.c. motor connected to a 12 V supply of negligible resistance. The coil of the motor has a constant resistance, and the only energy losses in the winch are due to heating in this coil. The winch raises loads vertically at constant speed; a heavier load makes the motor turn more slowly and draw a larger current.

When it raises a load of mass 0.50 kg, the current is 1.0 A and the load rises at 2.0 m s^{-1}. When it raises a load of mass 1.5 kg, the current is 3.0 A. Take $g = 10$ N kg^{-1}.

At what speed does the 1.5 kg load rise, and what would the current be if the drum of the winch were clamped so that it could not turn?

A 1.2 m s^{-1}; zero

B 1.2 m s^{-1}; 6.0 A

C 2.0 m s^{-1}; 1.0 A

D 0.67 m s^{-1}; 6.0 A

E 2.0 m s^{-1}; 6.0 A

F 1.2 m s^{-1}; 1.0 A

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8. Sports scientists test the reactive strength of an athlete's legs with a drop jump. The athlete, of mass 60 kg, steps off a box so that her centre of mass falls freely from rest through 0.45 m before her feet touch a force plate. She is in contact with the plate for 0.25 s and then springs vertically upwards, leaving the plate with her centre of mass at the same height as when she landed.

The *reactive strength index* (RSI) of a jump is defined as the height, in metres, through which the centre of mass rises after leaving the plate, divided by the contact time, in seconds. For this jump the RSI is 3.2.

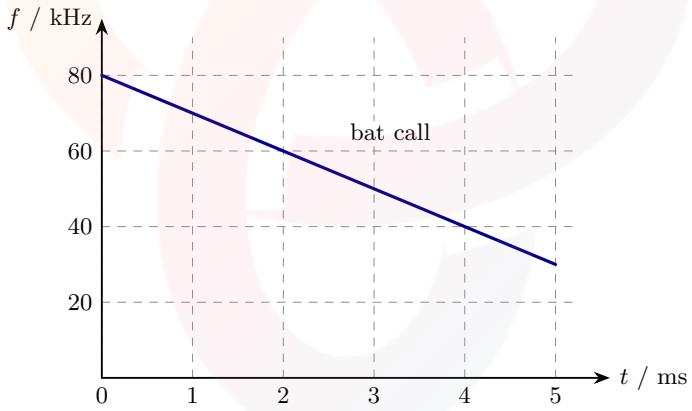
Ignore air resistance and take $g = 10$ N kg^{-1}. What is the mean force exerted on the athlete by the force plate while she is in contact with it?

A 1680 N **B** 2280 N **C** 840 N

D 1320 N **E** 1080 N **F** 1560 N

9. A bat detector lets people listen to the ultrasonic calls of bats. It is tuned to a chosen frequency f_0. Whenever its microphone receives a sound of frequency f, its loudspeaker plays a sound whose frequency is the difference between f and f_0 (the larger minus the smaller).

A bat makes a call lasting 5.0 ms, during which the frequency of the call falls steadily from 80 kHz to 30 kHz, as shown in the graph. The detector is tuned to $f_0 = 45$ kHz, and the listener can hear sounds with frequencies from 20 Hz to 20 kHz. Ignore the very short times during which the loudspeaker's frequency is below 20 Hz.



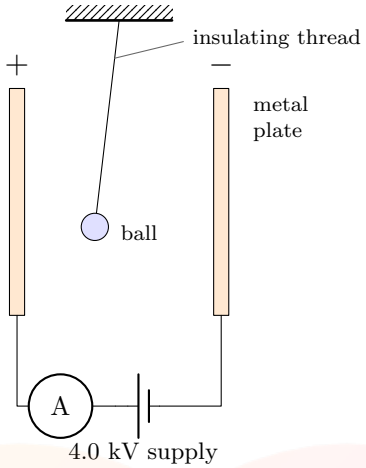
For how long during the call does the listener hear a sound, and what frequency does she hear at the end of the call?

A 3.5 ms; 15 kHz **B** 3.5 ms; 35 kHz **C** 5.0 ms; 15 kHz

D 4.0 ms; 15 kHz **E** 1.5 ms; 15 kHz **F** 2.0 ms; no sound is heard

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10. In a demonstration, two vertical metal plates face each other a few centimetres apart and are connected to a 4.0 kV d.c. supply. A sensitive ammeter is connected in the lead between the supply and the positive plate. A light ball with a conducting surface hangs between the plates from an insulating thread, as shown.



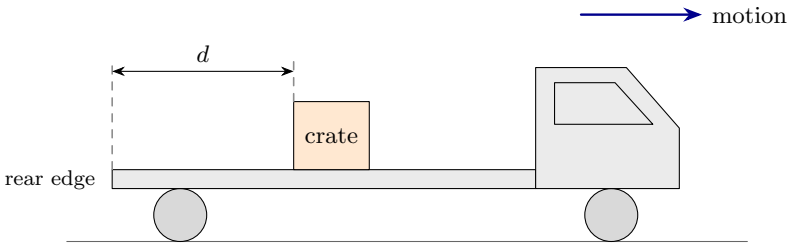
Once set going, the ball swings to and fro, touching each plate in turn. Each time the ball touches a plate, its charge changes to a charge of magnitude q (the same every time) with the same sign as that plate, and the charge on the ball does not change while it crosses the gap. No charge leaks away through the air.

The ball touches the negative plate 5.0 times each second, and the ammeter shows a steady average reading of 40 nA. What is q, and how much work does the electric field do on the ball during a single crossing of the gap?

A 4.0 nC; 8.0 μJ **B** 2.0 nC; 8.0 μJ **C** 4.0 nC; 16 μJ

D 8.0 nC; 32 μJ **E** 8.0 nC; 16 μJ **F** 4.0 nC; 32 μJ

11. A crate stands on the flat, horizontal load bed of a truck, a distance d in front of the open rear edge. The truck is at rest. At $t = 0$ it moves off along a straight, level road with a constant acceleration of 4.0 m s^{-2} until $t = 3.0$ s, and then continues at a constant speed.



The friction force between the crate and the load bed can be no greater than 0.30 times the weight of the crate, and whenever the crate is sliding over the bed the friction force is equal to 0.30 times its weight. Treat the crate as small, so that it falls off only if it would move past the rear edge. Ignore air resistance and take $g = 10$ N kg^{-1}.

What is the smallest value of d for which the crate does not fall off the truck?

A 6.0 m **B** 4.5 m **C** 24 m

D 12 m **E** 18 m **F** 1.5 m

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12. A biologist uses a simple model to compare how animals of different sizes jump. Animals A and B have exactly the same shape and are made of the same materials, but every length of B is twice the corresponding length of A.

In the model, the force that a muscle can exert is proportional to its cross-sectional area. Each animal jumps vertically from a crouch by pushing on the ground with a constant force, during which its centre of mass rises through a distance proportional to the animal's length, and it then leaves the ground and moves freely under gravity. During the push, the animal's weight may be ignored compared with the push force. Ignore air resistance.

According to this model, which of the following statements is/are correct?

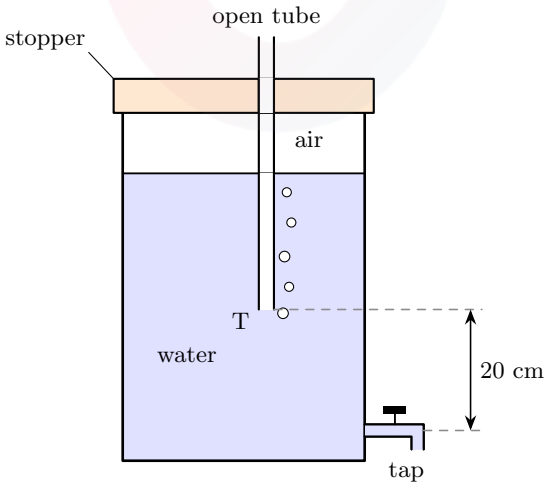
I. A and B leave the ground at the same speed.

II. The push of B lasts twice as long as the push of A.

III. During the push, the mean power developed by B is 8 times the mean power developed by A.

A III only
B I, II and III
C II only
D I and III only
E II and III only
F I and II only
G none of them
H I only

13. The figure shows a bottle designed to deliver water at a steady rate. The rigid bottle is sealed at the top by an airtight stopper, through which passes an open vertical tube. The lower end T of the tube is under the water. Near the bottom of the bottle is a small tap, 20 cm below T.



The tap is opened. Water drips out slowly, the air above the water expands, and the water level inside the tube falls until it reaches T. From then on, while the water surface in the bottle is still above T, air enters as a slow stream of small bubbles from T, and the tube contains only air. Treat the water in the bottle as being at rest, and ignore surface tension and the weight of the air in the tube. Take $g = 10$ N kg^{-1} and the density of water as 1000 kg m^{-3}.

Consider the following statements about the time during which bubbles are entering.

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I. The pressure in the water at the level of the tap is 2.0 kPa above atmospheric pressure, and it stays the same as the water surface falls towards T.

II. When the water surface is 15 cm above T, the pressure of the air above the water is 1.5 kPa below atmospheric pressure.

III. As the water surface falls towards T, the pressure of the air above the water decreases, because this air is expanding.

IV. If the tube were pushed further down through the stopper, so that T was lower but still above the tap, then once bubbles were again entering steadily, the pressure in the water at the level of the tap would be lower than before.

Which of these statements must be true?

A I only

B I, II, III and IV

C I, III and IV only

D I, II and IV only

E II only

F I and II only

G I and III only

H II and III only

14. In a hospital, liquid waste containing a radioactive isotope is stored in a shielded tank. The isotope has a half-life of 6.0 hours, and its decay product is stable. The tank starts empty, and every 12.0 hours a batch of waste is added; every batch contains the same number of nuclei of the isotope. Nothing leaves the tank.

A detector fixed beside the tank registers a steady background count rate of 50 counts per minute, together with a count rate from the tank that is proportional to the activity of the isotope in the tank. After many days, the reading just after each batch is added has settled at 850 counts per minute.

What was the reading just after the first batch was added, and just after which batch did the reading first exceed 840 counts per minute?

A 450 counts per minute; the seventh

B 600 counts per minute; the third

C 650 counts per minute; the fourth

D 650 counts per minute; the third

E 250 counts per minute; the fourth

F 650 counts per minute; the fifth

G 600 counts per minute; the fourth

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15. An ultrasonic anemometer measures the speed of the air by timing ultrasound pulses. Two transducers, P and Q, are fixed 16.8 cm apart, and each can both send and receive short pulses. The instrument measures the time a pulse takes to travel from P to Q, and the time a pulse takes to travel from Q to P. Sound travels at the same speed relative to the air in every direction, and this speed depends only on the air temperature θ:

$$v = 331 + 0.60\,\theta,$$

with v in m s^{-1} and θ in °C.

The anemometer is mounted on a motor yacht that travels due north in a straight line at a steady 6.0 m s^{-1} relative to the water, which is still. The line PQ points north-south, with P at the northern end, and the air moves horizontally along the north-south line. A pulse takes 480 μs to travel from P to Q, and 500 μs to travel from Q to P.

What is the velocity of the air relative to the water, and what is the air temperature?

A 13 m s^{-1} towards the south; 20 °C

B 1.0 m s^{-1} towards the south; 32 °C

C 7.0 m s^{-1} towards the south; 20 °C

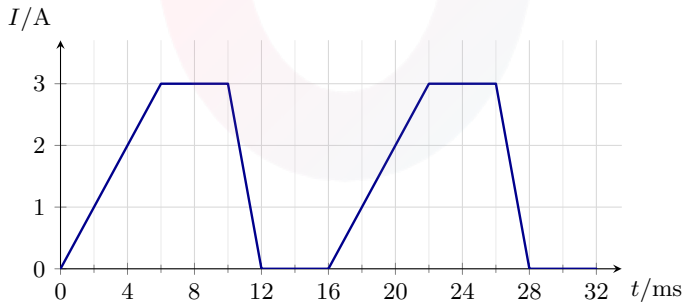
D 1.0 m s^{-1} towards the south; 20 °C

E 1.0 m s^{-1} towards the north; 20 °C

F 13 m s^{-1} towards the north; 20 °C

G 1.0 m s^{-1} towards the south; 8 °C

16. Two coils, 1 and 2, are wound on the same closed soft iron core, as in an ignition coil. A generator drives a current I through coil 1 that varies with time t as shown; the pattern repeats every 16 ms.



The ends of coil 2 are connected to an 8.0 Ω resistor and nothing else. At every instant, the p.d. induced across coil 2 is proportional to the rate of change of I: when I changes at a steady 1.0 A per millisecond, this p.d. is 4.0 V. Coil 2 has negligible resistance, and the current in coil 2 has no effect on I.

What is the mean power dissipated in the resistor?

A 0.75 W

B 2.5 W

C 0.19 W

D 1.5 W

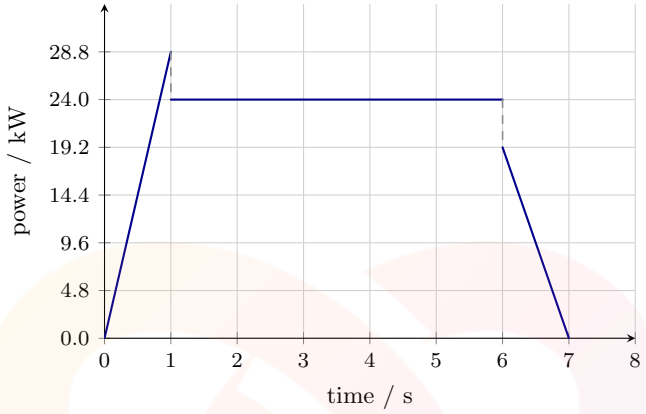
E 0.56 W

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F 0.28 W

G 0.25 W

17. A winch on a test rig hoists a load vertically, starting and finishing at rest. The cable pulls on the load with one constant force while the load speeds up, then with a different constant force while it moves at constant speed, and then with a third constant force while it slows down. The graph shows the power delivered to the load by the cable (the rate at which the cable does work on the load) against time.

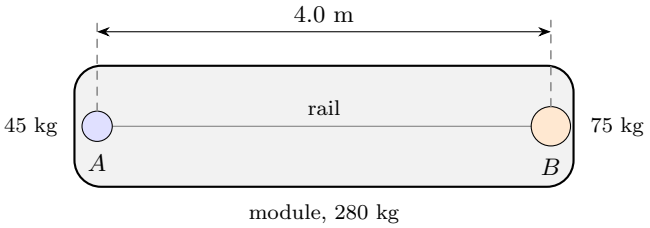


Ignore air resistance and the mass of the cable, and take $g = 10$ N kg^{-1}. What are the mass of the load and the total height through which it is raised?

A 1440 kg and 12 m **B** 2400 kg and 6.0 m **C** 1200 kg and 12 m

D 1200 kg and 14 m **E** 1200 kg and 10 m **F** 1440 kg and 10 m

18. A cylindrical module floats freely in deep space, far from any other object, and is at rest. Its mass is 280 kg. Astronaut A, of mass 45 kg, is at rest at one end of the module, and astronaut B, of mass 75 kg, is at rest at the other end, 4.0 m away along the axis. They swap places by pulling themselves along a rail on the axis of the module, so that each ends at rest at the point inside the module where the other started.



Treat each astronaut as a particle moving along the axis. When both astronauts are again at rest, how far has the module moved from its original position, and in which direction?

A It has not moved, because no external force acts.

B 0.60 m, in the direction in which A moved

C 0.43 m, in the direction in which A moved

D 0.30 m, in the direction in which B moved

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E 1.0 m, in the direction in which A moved

F 0.30 m, in the direction in which A moved

19. Water from a boiler flows through a radiator in a room that is kept at 20°C. The water enters the radiator at 60°C and, with the pump on its normal setting, leaves at 40°C. The radiator then gives out energy to the room at 1.5 kW.

Use the following model. The rate at which the radiator gives out energy is proportional to the difference between its mean temperature and the room temperature, where its mean temperature is the average of the inlet and outlet water temperatures. All the energy given out comes from the water flowing through, and conditions are steady.

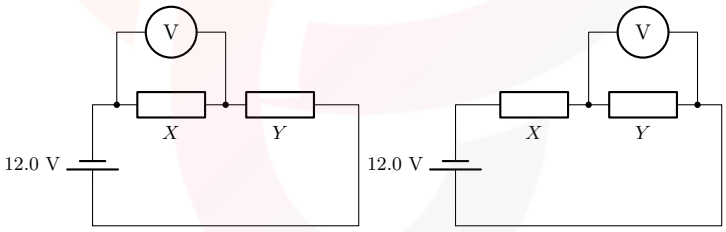
The pump is turned down so that the mass of water flowing through the radiator each second is halved. The water still enters at 60°C and the room stays at 20°C. Once conditions are steady again, at what rate does the radiator give out energy?

A 1290 W **B** 1200 W **C** 750 W

D 2400 W **E** 1500 W **F** 1000 W

20. A technician checks a circuit in which two resistors, X and Y, are connected in series to a 12.0 V battery of negligible internal resistance. Neither resistance is known. She has only one voltmeter, and its resistance, which is also unknown, is not large enough to ignore.

With the voltmeter connected across X, it reads 4.0 V. With the same voltmeter connected across Y instead, it reads 6.0 V.



(a) reading 4.0 V (b) reading 6.0 V

What is the p.d. across X when the voltmeter is not connected, and what is the resistance of the voltmeter as a multiple of the resistance of X?

A 5.0 V; 3

B 4.8 V; 6

C 4.8 V; 2

D 4.8 V; 3

E 4.8 V; 1.5

F 4.0 V; 2

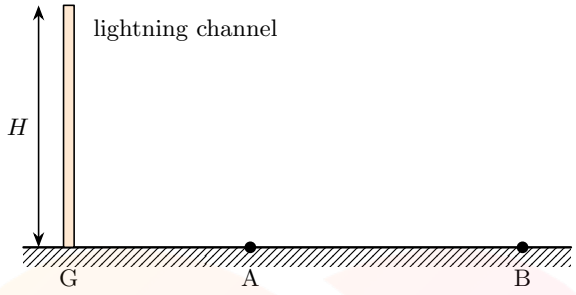
G 5.0 V; 2

H 4.0 V; 3

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21. A lightning stroke can be modelled as a straight vertical channel rising from a point G on flat ground to a height H. At the instant of the stroke, thunder is emitted at the same moment from every point of the channel, and it spreads out in all directions at 340 m s^{-1}. Ignore wind and reflections.

Two microphones, A and B, stand on the flat ground on a straight line through G, on the same side of G, with B further from G than A. They are connected to the same data logger, but nothing records the flash. Measuring time t from when the logger was switched on, microphone A picks up thunder from $t = 10.0$ s until $t = 12.0$ s, and microphone B picks up thunder from $t = 14.5$ s until $t = 15.5$ s.



What is the height H of the channel?

A 1.02 km

B 1.36 km

C 2.26 km

D 1.70 km

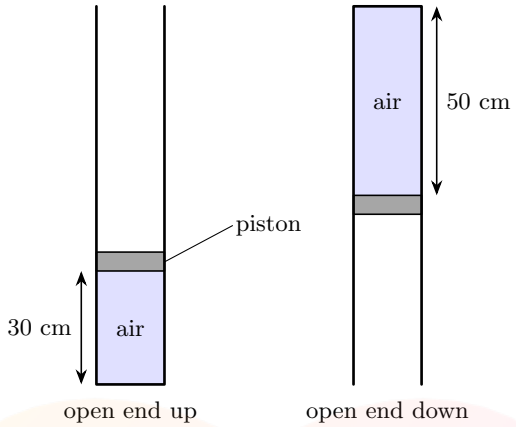
E 0.68 km

F 2.55 km

G 1.53 km

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22. A narrow cylinder of uniform cross-section, 1.0 m long, is closed at one end. Some air is sealed inside it by a piston, which fits airtightly and slides without friction. The length of the air column is the distance from the closed end to the piston. When the cylinder is held vertically at rest with its open end upwards, the air column is 30 cm long. When it is held vertically at rest with its open end downwards, the air column is 50 cm long.



The cylinder is then fixed vertically, open end downwards, inside a lift. The lift accelerates upwards with a constant acceleration of 5.0 m s^{-2}, and the piston settles in a new position relative to the cylinder. The temperature of the air stays constant, the atmospheric pressure is the same throughout, and $g = 10$ N kg^{-1}.

What is the length of the air column while the lift is accelerating?

A 27.3 cm

B 60 cm

C 42.9 cm

D 37.5 cm

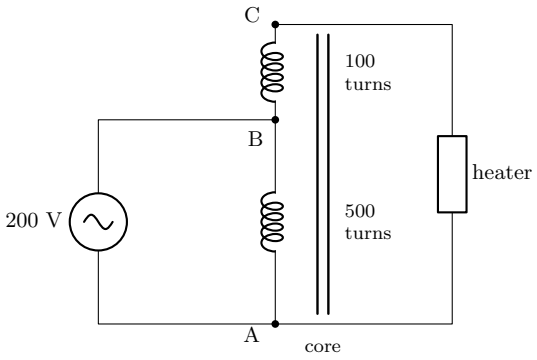
E 75 cm

F 50 cm

G 80 cm

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23. At the end of a long rural supply line, the mains p.d. has fallen to 200 V. A workshop raises it with a booster transformer that has only one coil. The coil, of 600 turns, is wound on a closed soft iron core. It has terminals A and C at its ends and a tapping point B, which is 500 turns from A and 100 turns from C. The supply is connected between A and B, and a heater is connected between A and C, as shown.



Treat the transformer as ideal: at every instant the p.d. across any section of the coil is proportional to its number of turns, and the power delivered to the heater equals the power taken from the supply. All p.d.s and currents are effective values, which may be used exactly as for direct current. The heater behaves as a 24 Ω resistor.

The coil is wound from the same wire throughout, so the resistance of each section is proportional to its number of turns. This resistance is small enough to ignore when finding the p.d.s and currents, but the wire is still warmed by the currents in it. What is the ratio

rate of heating in section AB : rate of heating in section BC ?

A 36 : 5

B 1 : 25

C 121 : 5

D 5 : 36

E 125 : 1

F 1 : 5

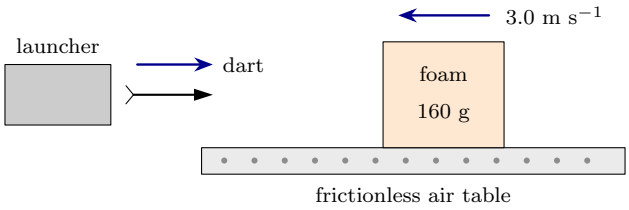
G 5 : 1

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24. In a test of protective packaging, a steel dart of mass 90 g is fired horizontally into a foam block of mass 160 g, along a line through the centre of the block. While the dart is moving through the foam, the foam exerts a constant resistive force on it, of the same size in every test, and the dart exerts an equal and opposite force on the foam. The dart moves in a straight line, and no material is lost.

In Test 1 the block is clamped so that it cannot move. The dart only just passes right through the block when it is fired at 8.0 m s^{-1}.

In Test 2 an identical block rests on a frictionless air table and is set sliding towards the launcher at 3.0 m s^{-1}. The dart is fired along the line of the block's motion.



What is the smallest speed, relative to the ground, at which the dart must be fired in Test 2 for it to pass right through the block?

A 8.0 m s^{-1}

B 9.5 m s^{-1}

C 11 m s^{-1}

D 10 m s^{-1}

E 7.0 m s^{-1}

F 13 m s^{-1}

G 5.0 m s^{-1}

25. To calibrate radon detectors, a sealed glass ampoule of fixed volume is filled with a mixture of nitrogen and radon-222, $^{222}_{86}$Rn, a gas with a half-life of 3.8 days. The ampoule is kept at a constant temperature.

Each radon nucleus that decays is followed, within a few hours, by a series of further alpha and beta-minus decays ending at lead-210, $^{210}_{82}$Pb. Treat these further decays as instantaneous, and treat lead-210 as stable over the times involved. The lead and the short-lived nuclides in between are solids, which stick to the glass and have negligible volume. The beta particles are absorbed by the glass, and each alpha particle quickly gains two electrons to become a helium atom, which stays in the ampoule as a gas. No chemical reactions occur. At constant temperature, the pressure of a gas mixture in a fixed volume is proportional to the total number of gas particles (atoms or molecules) that it contains.

The pressure in the ampoule was 100 kPa when it was sealed, and 120 kPa exactly 3.8 days later. What fraction of the gas particles in the ampoule were radon atoms when it was sealed, and what will the pressure be 11.4 days after sealing?

A $\frac{1}{5}$; 135 kPa

B $\frac{2}{5}$; 135 kPa

C $\frac{1}{10}$; 135 kPa

D $\frac{1}{5}$; 140 kPa

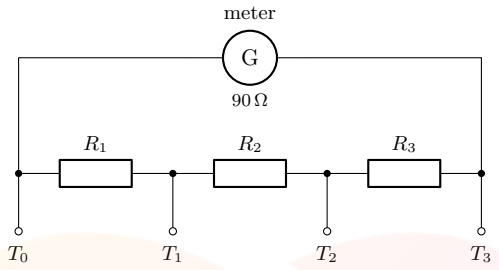
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E $\frac{1}{5}$; 160 kPa

F $\frac{1}{5}$; 130 kPa

G $\frac{2}{15}$; 135 kPa

26. A moving coil meter has a resistance of 90 Ω, and its needle reaches the end of its scale when the current through it is 1.0 mA. It is made into an ammeter with three ranges by connecting three resistors R_1, R_2 and R_3 in series as a chain, with the meter connected across the whole chain, as shown.



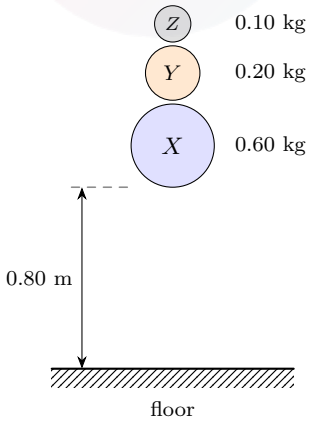
The ammeter is connected into a circuit through the common terminal T_0 and just one of the terminals T_1, T_2 and T_3; the other two are left unconnected. The resistors are chosen so that the needle reaches the end of the scale when the current entering the ammeter is 1.0 A through T_1, 100 mA through T_2 and 10 mA through T_3.

What is the resistance of R_2?

A 0.90 Ω **B** 0.82 Ω **C** 1.0 Ω

D 0.91 Ω **E** 0.10 Ω **F** 9.0 Ω

27. In a lecture demonstration, three balls are held in a vertical stack, each just above the one below, and released together from rest with the bottom of the stack 0.80 m above a hard floor. The bottom ball X has mass 0.60 kg, the middle ball Y has mass 0.20 kg and the top ball Z has mass 0.10 kg.



Treat the balls as particles that each fall 0.80 m before X reaches the floor. X bounces off the floor, then immediately collides with Y, which then immediately collides with Z. Every collision is elastic (no kinetic energy is lost), head-on and vertical, and so brief that the effect of gravity during it can be ignored, and each collision involves only two objects. Ignore air resistance and take $g = 10$ m s^{-2}.

To what greatest height above the floor does Z rise?

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A 14.4 m

B 12.8 m

C 6.4 m

D 39.2 m

E 7.2 m

F 3.2 m

G 0.80 m

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